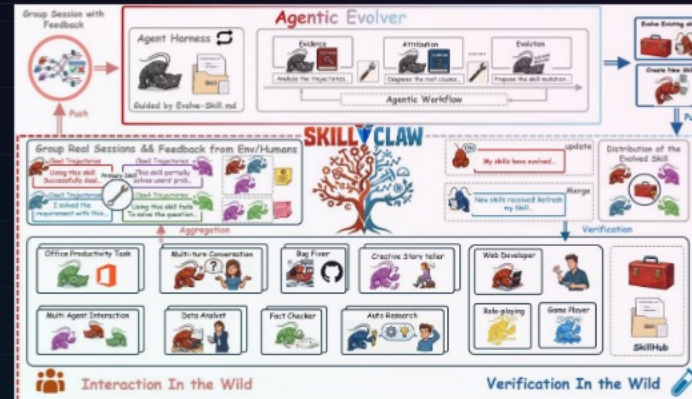




SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



01

Evidence

Analyze trajectories to identify behavioral patterns across sessions

02

Attribution

Diagnose root causes of failures and successes per skill

03

Evolution

Propose skill mutations — refine existing or create new skills

04

Distribution

Verify, merge into shared repo, and sync back to all agents

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

1

Evidence

Analyze structured session trajectories preserving full action–feedback causal chains. Aggregate across users and group by referenced skills to form a shared evidence base.

2

Attribution

Diagnose root causes of failures and successes. Different users exercising the same skill under diverse contexts produce complementary views of behavioral boundaries.

Open-ended reasoning: The evolver is an autonomous agent performing reasoning over evidence — not a set of hardcoded heuristics. This enables context-aware mutations that rule-based systems cannot produce.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

3 of 6 days yielded accepted updates – successful iterative refinement before plateauing

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

3 of 6 days yielded accepted updates, showing successful iterative refinement before reaching a plateau.

Key Takeaways: Three Properties That Distinguish SkillClaw

01

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem — not isolated per-user gains.

02

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation, explicit user intervention, or predefined update rules.

